MMB-117

Amplicon sequence data analysis

Workflow wrap-up

- Pre-processing
- DADA2 pipeline
- Data exploration
- Statistical analysis

Further reading

- R packages for microbiome data analysis:
  - coda4microbiome (https://malucalle.github.io/coda4microbiome)
  - miaverse (https://microbiome.github.io/)
- Data analysis tutorials:
  - Happy Belly Bioinformatics (https://astrobiomike.github.io/amplicon/)
  - Pat Schloss YouTube channel (https://www.youtube.com/@Riffomonas)
- Modern Statistics for Modern Biology (https://www.huber.embl.de/msmb/)
- Articles in Moodle

Learning outcomes revisited

- Understand the basics of amplicon sequencing and bioinformatic approaches to analyse amplicon sequencing data
- Be able to plan an amplicon sequencing project and choose the right tools and approaches to answer your specific research question
- Have confidence to learn new methods needed to answer your research question in microbial ecology
- Empower you to ask and answer the questions you have on your own data

Report instructions

- Max 10 pages (including references)
  - Font size 12 (references 10)
  - Line spacing 1.5
- Manuscript format
  - Introduction/background and aim(s)
  - Materials and methods
  - Results
  - Short discussion w/ some conclusions
  - References
- Deadline 31.3.2025 (Puhti project closes early April).

Q & A